```
openssl genrsa -des3 -rand /etc/hosts -out smtpd.key 1024
chmod 600 smtpd.key
```

The above command generates a private key using a triple DES cipher that uses pseudo-random bytes and writes it to a file *smtpd.key*.

Next, create a certificate using the key we just created. Execute the following command in a terminal:

```
openssl req -new -key smtpd.key -out smtpd.csr
```

You should see the following output:

```
Enter pass phrase for smtpd.key:

You are about to be asked to enter information that will be
incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished
Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [XX]:IN
State or Province Name (full name) []:Tamilnadu
Locality Name (eg, city) [Default City]:Chennai
Organization Name (eg, company) [Default Company Ltd]:Sastra
Technologies Pvt. Ltd.,
Organizational Unit Name (eg, section) []:Netraja
Common Name (eg, your name or your server's hostname)
[]:sastratechnologies.net
Email Address []:info@sastratechnologies.in
```

Please enter the following 'extra' attributes to be sent with your certificate request:

```
A challenge password []:
An optional company name []:
```

This will create a file *smtpd.csr*. Execute the following command on your terminal:

```
openssl x509 -req -days 365 -in smtpd.csr -signkey smtpd.key
-out smtpd.crt
```

You should see the following output:

```
Signature ok
subject=/C=IN/ST=Tamilnadu/L=Chennai/O=Sastra Technologies
Pvt. Ltd.,/OU=Netraja/CN=sastratechnologies.net/
emailAddress=info@sastratechnologies.in
Getting Private key
Enter pass phrase for smtpd.key:
This will create the certificate file smtpd.crt.
```

Now execute the following command:

```
openssl rsa -in smtpd.key -out smtpd.key.unencrypted
```

You should see the following output:

```
Enter pass phrase for smtpd.key:
writing RSA key
```

This will create the unencrypted key and write a file *smtpd.key.unencrypted*. Now move the unencrypted key to the *smtpd.key* file

```
mv -f smtpd.key.unencrypted smtpd.key
```

…and generate the RSA key:

```
openssl req -new -x509 -extensions v3_ca -keyout cakey.pem
-out cacert.pem -days 365
```

You should see the following output:

```
Generating a 2048 bit RSA private key
.....................................+++
.......................+++
writing new private key to 'cakey.pem'
Enter PEM pass phrase:
Verifying - Enter PEM pass phrase:
-----
You are about to be asked to enter information that will be
incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished
Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [XX]:IN
State or Province Name (full name) []:Tamilnadu
Locality Name (eg, city) [Default City]:Chennai
Organization Name (eg, company) [Default Company Ltd]:Sastra
Technologies Pvt. Ltd.,
Organizational Unit Name (eg, section) []:Netraja
Common Name (eg, your name or your server's hostname)
[]:sastratechnologies.net
Email Address []:info@sastratechnologies.in
```

Update the DNS Zone entries

After generating the SSL keys, set up the DNS Zone entries so